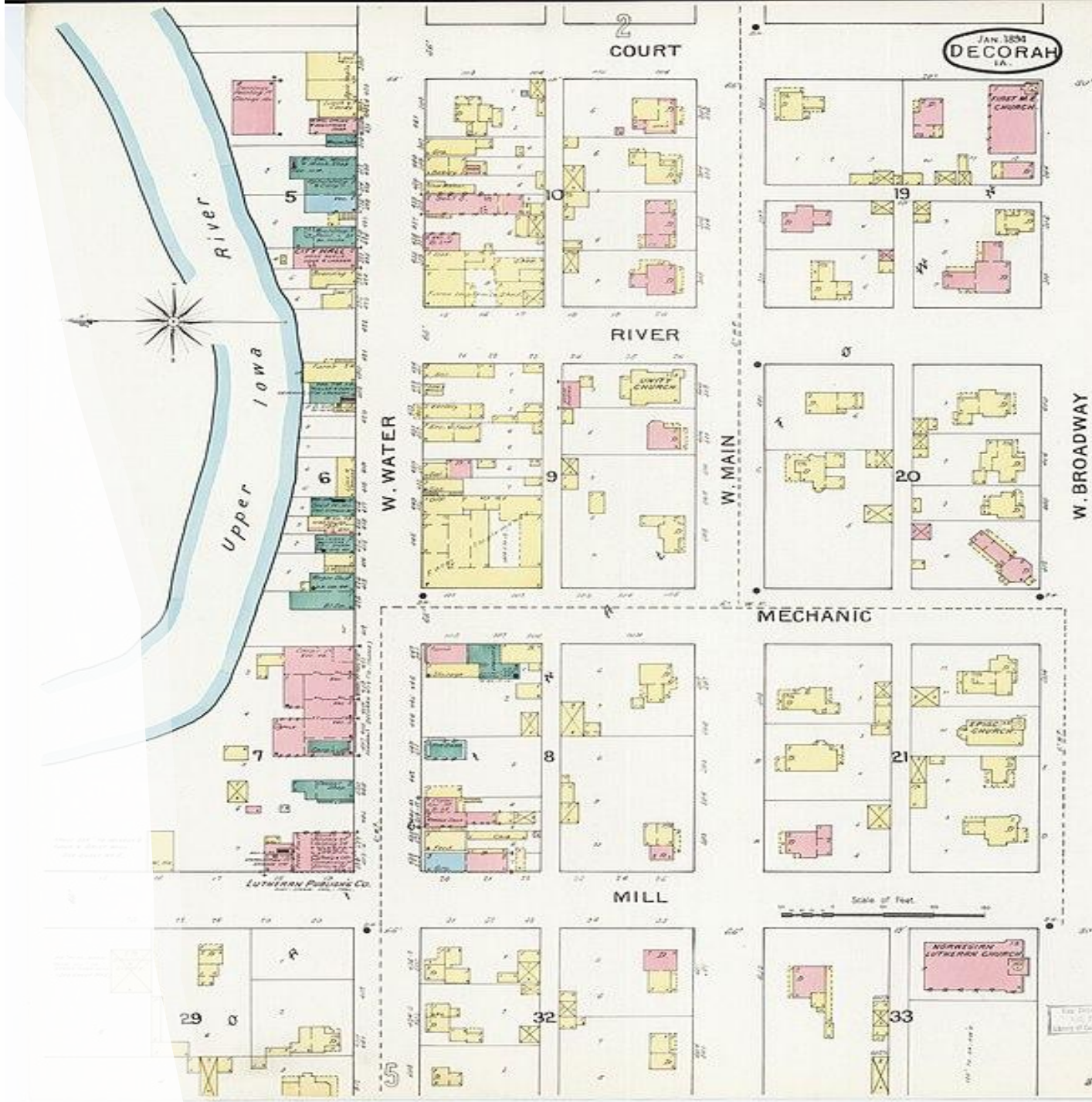
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Interpreting Hepatitis B Serological Laboratory Results

In Person Public Health Training

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Learning Objectives

- By the end of the training the participants will be able to define key hepatitis B testing terms and serologic lab markers with at least 80% accuracy on the post knowledge check. Assessment part one quiz.
- Given a patient history and hepatitis B laboratory results participants will be able to interpret a patient's case classification and defend their assessment correctly for at least 2/3 patients. Assessment part two group interpretation and history.
- Given the patient history and lab result the participants will be able to classify a patient's case status using the 2024 CDC case definitions and handout for at least 2/3 patients. Assessment part two case classification.

A patient’s laboratory result lands on your desk!

HBsAg +, Total Anti-HBc +, IgM Anti-HBc -, Anti-HBs –

What does it mean and what do you do next?

Key Definitions: Testing

- Screening Vs. Testing
  - Screening: testing people who are asymptomatic to find possible infection or immunity
  - Testing (diagnostic): testing people who are symptomatic, an identified exposure (e.g., sexual contact, injection drug use), or a prior positive result (e.g., HBsAg)
- Antigen Vs. Antibody
  - Antigen: a non-self molecule the immune system uses to identify pathogens (HBsAg, HBeAg)
  - Antibody: a protein the immune system makes to attack antigens-pathogens (Anti-HBs, Anti-HBe)

Key Definitions: Results

- Qualitative Vs. Quantitative
  - Qualitative: positive/negative, non/reactive
  - Quantitative: a numeric value (Anti-HBs >=10 mIU/mL, HBV DNA 2,000 IU/mL)
- Detectable Vs. Undetectable
  - Detectable: at or above the lab test's limit of detection
  - Undetectable: below the lab test's limit of detection, not proof of zero (HBV DNA below cutoff is not positive but not negative)

Viral Hepatitis Language

- **Acute** (short-term/new) **infection:**
  - When the infection is newly acquired.
  - Approximately 80% of patients will be asymptomatic during the acute phase of infection.
- **Chronic** (long-term/previous) **infection:**
  - Infection lasts 6 months or more and is usually life-long unless treated.
- **Resolved or cleared infection:**
  - The body getting rid of the hepatitis infection—usually in the acute stage.
  - Patients can spontaneously clear the virus or clear the virus through treatment.

Modes of Transmission

- **Primary** (most common)
  - Perinatal; vertical transmission from a mother who has hepatitis B to her baby
  - Sharing injection drug use equipment
  - Sexual contact (e.g., anal, vaginal)
- **Secondary** (less common)
  - Needle-sticks; sharps injuries
  - Razors, glucose monitoring equipment, toothbrushes*
    - *If blood to blood transmission is present
  - Organ transplant and dialysis

Source: Centers for Disease Control and Prevention. Viral hepatitis surveillance—United States, 2017. Atlanta: US Department of Health and Human Services, Centers for Disease Control and Prevention; 2019. Available at: https://www.cdc.gov/hepatitis/statistics/2017surveillance/index.htm.

Why test for hepatitis B?

- Screening, one-time adult triple panel: HBsAg, Anti-HBs, Total Anti-HBc
- Occupational screening
- Diagnosis: acute/chronic, confirmed/probable
- Immune status: immune or susceptible
- Perinatal screen
- Monitoring chronic infection
- Exposure to contact with hepatitis B

Core Serologic Laboratory Results

- HBsAg, surface antigen
  - Viral hepatitis surface antigen, present in acute and chronic patients, infectious
- Total IgM/IgG Anti-HBc, core antibody
  - Past (resolved) or ongoing (chronic) infection, lifelong
- IgM Anti-HBc
  - Recent infection within six months, indicative of acute infection
- Anti-HBs, surface antibody
  - Recovery and immunity from infection or successful vaccination

Additional Labs

- HBeAg and Anti-HBe
  - E-antigen active viral replication and antibody to the e-antigen
- HBV DNA
  - Genetic viral material, quantitative results for viral load
- ALT, AST, Bilirubin
  - Liver damage ALT>200 IU/L or bilirubin>=3.0mg/dL meet acute definition

Interpreting panels

HBsAg	Total Anti-HBc	IgM Anti-HBc	Anti-HBs	Interpretation
-	-	-	-	Never infected, susceptible, vaccinate
+	-	-	-	Early acute or false positive up to 18 days after vaccination
+	+	+	-	Acute infection
-	+	+	-	Acute infection, resolving
-	+	-	+	Recovered from past infection, immune
+	+	-	-	Chronic infection
-	+	-	-	Isolated core, past infection, low level chronic, or false positive
-	-	-	+	Immune from vaccination (Anti-HBs>=10mIU/mL)

Interpret together

Patient One, results: HBsAg -, Total Anti-HBc -, Anti-HBs + 120 mIU/mL

Patient Two, results: HBsAg +, Total Anti-HBc +, IgM Anti-HBc +

History Taking

- Timeline of symptoms (e.g., jaundice)
- Risk factors and exposures (e.g., injection drug use, sex)
- Demographics and country of origin
- Any prior hepatitis B laboratory results (e.g., negative or positive)
- Reason for testing

Notable Circumstances

- False positive HBsAg due to recent hepatitis B vaccination (most common)
- Perinatal hepatitis B and infants need HBIG if mother is hepatitis B positive
- Immunocompromised patients
- Reactivation events in chronic or resolved infection during immunosuppressive therapies.

Common Mistakes

- A single positive result sent to state, negatives not shared (follow-up required)
- Disagreeing results (e.g., HBsAg – but HBV DNA +)
- IgM Anti-HBc in a reactivating chronic patient may read acute without additional lab results and history
- Unconfirmed laboratory results (only positive labs are reportable)
- Low or equivocal results
- No vaccination record

CDC 2024 Case Definitions

Classification	Laboratory Evidence	Clinical Criteria
Acute, Confirmed Tier 1	Positive HBsAg, HBeAg, or HBV DNA with IgM Anti-HBc or seroconversion within 12 months of negative HBsAg	No
Acute, Confirmed Tier 2	Positive HBsAg or HBV DNA with IgM Anti-HBc not done	Yes
Acute, Probable	Positive IgM Anti-HBc with HBsAg, HBeAg and HBV DNA negative or not done	Yes
Chronic, Confirmed	Positive HBsAg/HBeAg twice six months apart or HBsAg/HBeAg plus Total IgM/IgG Anti-HBc or HBsAg plus HBeAg or HBV DNA	No
Chronic, Probable	HBsAg or HBeAg with negative IgM Anti-HBc or not done	No

After Interpretation, Classification

- Linkage to care or ensure primary care provider
- Vaccinate any susceptible patients
- Contact tracing: any household, sexual, or injection drug use contacts
- Patient and provider follow-up

Group Case Review!

- Work in teams, review three patients' history and laboratory results
- For each patient complete the following:
  - Interpret laboratory results
  - Assess relevant history
  - Classify cases
  - Identify next steps
- You may use your handout

Recap and review assessment

- Review objectives
- Post knowledge check: 10 questions, no handout
- Group case review scored with the attached rubric